

Car Price Prediction Using Machine Learning Techniques

1. INTRODUCTION

Buying or selling a car usually depends on knowing the correct price of the vehicle. However, car prices can change a lot based on factors such as the brand, engine size, horsepower, fuel type, mileage, and many other features. Because of this, it is often difficult for customers and sellers to estimate the right price of a car. Traditional pricing methods depend on personal judgment or experience, which may not always be accurate.

With the growth of technology and data availability, machine learning has become an effective way to predict car prices. Machine learning models can study thousands of car features, identify patterns, and then produce accurate price predictions. This helps people make better decisions and reduces the chances of overpricing or underpricing a car.

This project focuses on predicting car prices using machine learning techniques. The main idea is to analyze how different factors—especially the brand of the car and other technical features—affect the price. By using past data, the system learns the relationship between car features and price, and then predicts the price of a new car with similar features.

In this study, three machine learning models are used: Linear Regression, Random Forest Regression, and XGBoost Regression. Their performances are compared to find out which model predicts car prices most accurately. The results show that machine learning can be a powerful and reliable tool for pricing cars, and such systems can be used by online car sellers, buyers, and automobile businesses.

2. PROBLEM STATEMENT

Car prices depend on many factors such as brand, engine power, fuel type, mileage, and vehicle dimensions. Because of these multiple factors, it becomes difficult for buyers and sellers to estimate the correct market price of a car. Traditional pricing methods rely on personal judgment or manual evaluation, which can be slow, inaccurate, and inconsistent.

There is a need for a data-driven system that can automatically analyze important car features and predict the price accurately. The problem addressed in this project is:

“To develop a machine learning model that can accurately predict the price of a car based on its brand and technical specifications.”

This model should help users understand how different features affect the price and provide a reliable prediction that reduces human error and improves decision-making.

3. OBJECTIVES OF THE STUDY

The main purpose of this project is to predict the price of a car using machine learning. To achieve this, the project has the following objectives:

1. To understand which car features affect the price.
2. To clean and prepare the dataset for analysis.
3. To analyze the data to find patterns and trends.
4. To create new useful features to improve prediction accuracy.
5. To apply machine learning models to predict car prices.

Models used:

- Linear Regression
 - Random Forest
 - XGBoost
6. To compare the accuracy of these models.
 7. To select the best model for final price prediction.

4. METHODOLOGY

The methodology of the project follows a structured machine learning pipeline.

Project Workflow

1. Data Collection
2. Data Cleaning
3. Feature Engineering

4. Exploratory Data Analysis
5. Train–Test Split
6. Model Building
7. Evaluation & Comparison
8. Final Model Deployment

Tools & Technologies

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- XGBoost library

Summary

The methodology followed a systematic machine learning workflow from data preparation to final model selection. This ensured reliable and accurate car price prediction results.

5. DATASET DESCRIPTION

The dataset used in this study is Car Price, which contains comprehensive information about various car models along with their technical specifications and market prices. The dataset is commonly used for automobile price prediction research and is well-suited for supervised machine learning tasks.

Dataset Size

- **Total Records:** 205 cars
- **Total Features:** 26 attributes
- **Target Variable:** Price

The dataset includes both numerical and categorical variables, allowing for detailed analysis of the factors that influence car prices.

Attributes in the Dataset

The major categories of features present in the dataset are as follows:

A. Identification Features

- **CarName:** Full name of the car, including brand and model.

B. Technical Specifications

These features describe engine performance and mechanical characteristics:

- enginesize
- horsepower
- stroke
- compressionratio
- peakrpm
- cylindernumber
- enginetype
- fuelsystem

These attributes directly affect performance and therefore influence the market price.

C. Physical Dimensions

These describe the overall size and structure of the vehicle:

- carlength
- carwidth
- carheight
- wheelbase
- curbweight

Vehicle dimensions contribute to comfort, build quality, and manufacturing cost.

D. Fuel and Drive Configuration

- fueltype (gas or diesel)
- aspiration (standard or turbo)
- carbody (sedan, hatchback, wagon, etc.)
- drivewheel (FWD, RWD, or 4WD)

These affect the design, efficiency, and drivetrain mechanism of the vehicle.

Target Variable

Price

- The selling price of the car in USD.
- This is the dependent variable the machine learning models aim to predict.

Data Quality Observations

- No major missing values were found.
- Some brand names contained spelling errors, which were corrected.
- The dataset included a mix of low, mid-range, and high-end (luxury) cars.
- The data was found suitable for regression analysis.

Reason for Dataset Selection

The dataset is suitable for this project because:

- It contains all major factors influencing car pricing.
- It provides rich variation across brands and models.
- It is ideal for training regression-based machine learning models.
- The size (205 rows) is manageable for academic experimentation and analysis.

6. Feature Engineering

In this project, feature engineering was done to improve the quality of the dataset and help the machine learning models make better predictions. First, two new features were created: **power_index**, which is calculated by multiplying horsepower and engine size to represent the overall engine performance, and **luxury_score**, which is obtained by adding the engine stroke and compression ratio to show the level of engine refinement. These new features help the model understand how engine strength and quality affect the car's price. Next, the original *CarName* column was removed because the important part, the car brand, was already extracted and corrected for spelling mistakes. All categorical features such as brand, fuel type, car body, and engine type were converted into numerical form using one-hot encoding so that the models

could understand them. Finally, only numerical columns were scaled using StandardScaler to keep all values on a similar range, which helps models like Linear Regression perform better. Overall, these feature engineering steps improved the dataset and made the price prediction models more accurate.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis helps us understand the dataset before building machine learning models. In this project, EDA was divided into three parts: Univariate, Bivariate, and Multivariate analysis. Different charts were used to study the distribution of values, relationships between features, and how they affect the car price.

7.1 Univariate Analysis

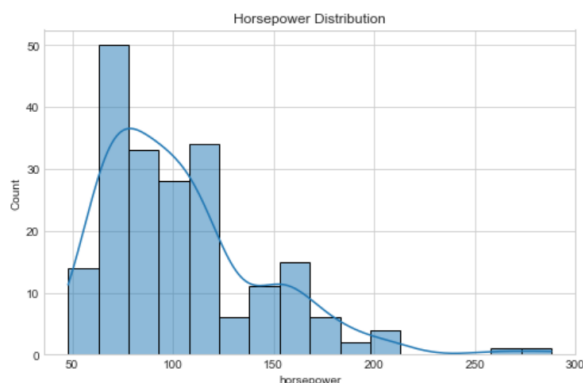
7.1.1 Price Distribution Chart



This chart shows how the car prices are spread out.

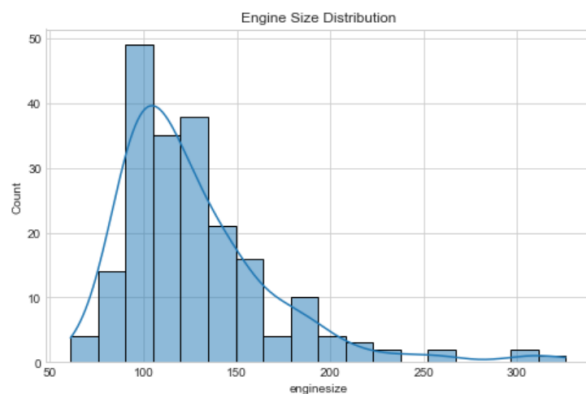
Most cars fall in the lower and mid-price ranges, with a few expensive luxury cars on the higher end.

7.1.2 Horsepower Distribution



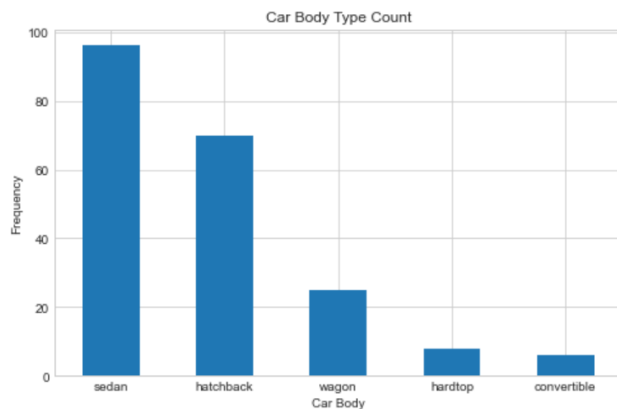
This graph shows the horsepower of the cars. Most cars have horsepower between 70 and 120. Only a few cars have very high horsepower above 150. This means most cars are normal-performance cars, and high-performance cars are fewer.

7.1.3 Engine Size Distribution



This graph shows the size of the car engines. Most cars have engine sizes between 90 and 130. Large engines are less common. So, small and medium engines are more popular in the dataset.

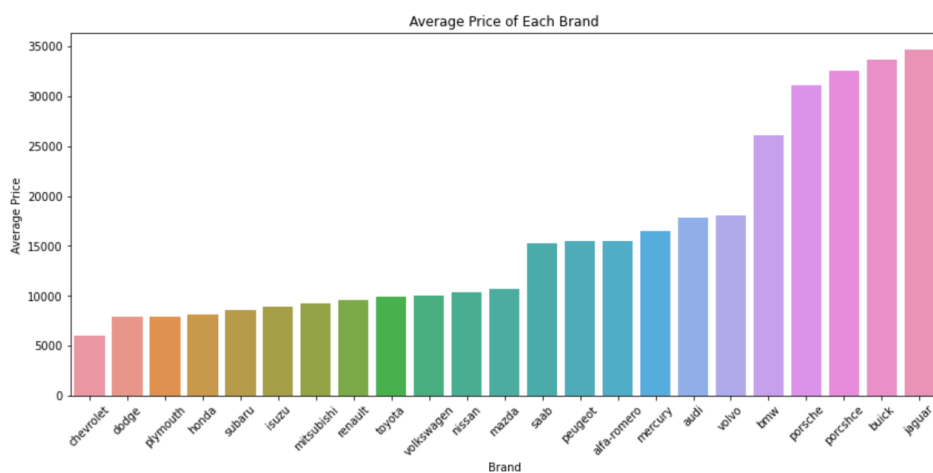
7.1.4 Car Body Type Count



This chart shows how many cars belong to each body type. Sedan is the most common, followed by hatchback. Wagon, hardtop, and convertible cars are much fewer. This means the dataset mostly contains regular day-to-day car types.

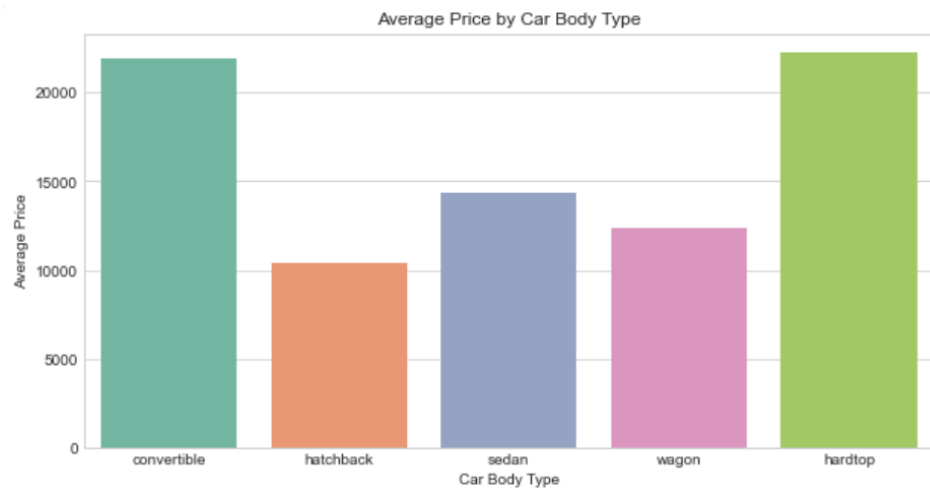
7.2 Bivariate Analysis

7.2.1 Average Price of Each Brand



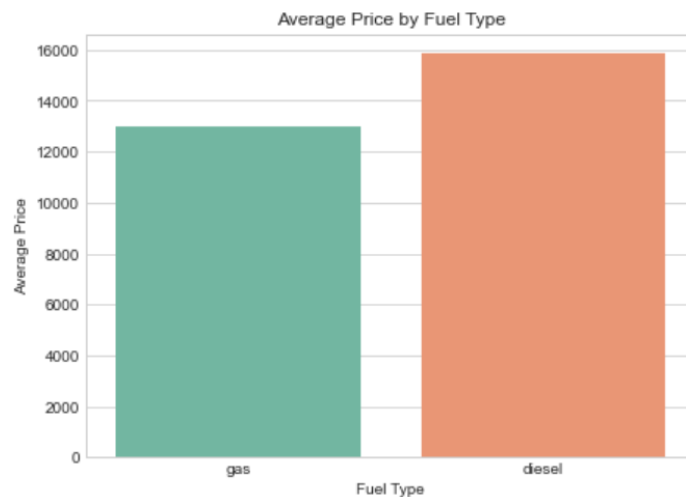
This bar chart shows the average car price for every brand in the dataset. We can clearly see that luxury brands like Jaguar, Porsche, BMW, and Buick have much higher prices compared to other brands.

7.2.2 Average Price by Car Body Type



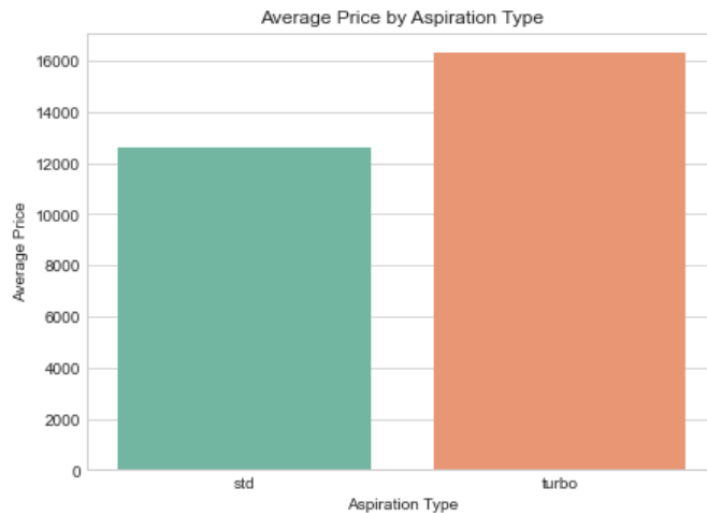
This chart compares the average price across different car body types. The results show that, Convertibles and hardtops are the most expensive body types, Sedans and wagons have moderate prices and Hatchbacks are usually the cheapest. This means the design and body style of a car also influence its price.

7.2.3 Average Price by Fuel Type



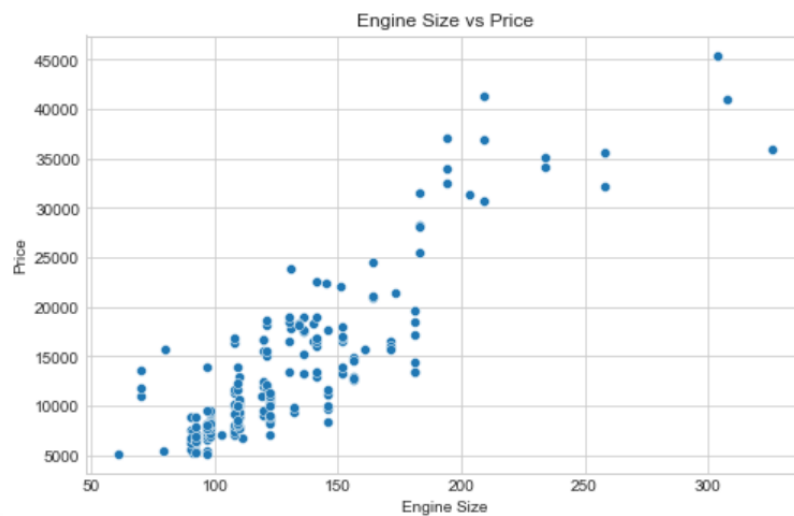
This plot shows how price differs between gas and diesel cars. Diesel cars have a higher average price than gas cars. This may be because diesel engines often offer better fuel efficiency and durability.

7.2.4 Average Price by Aspiration Type



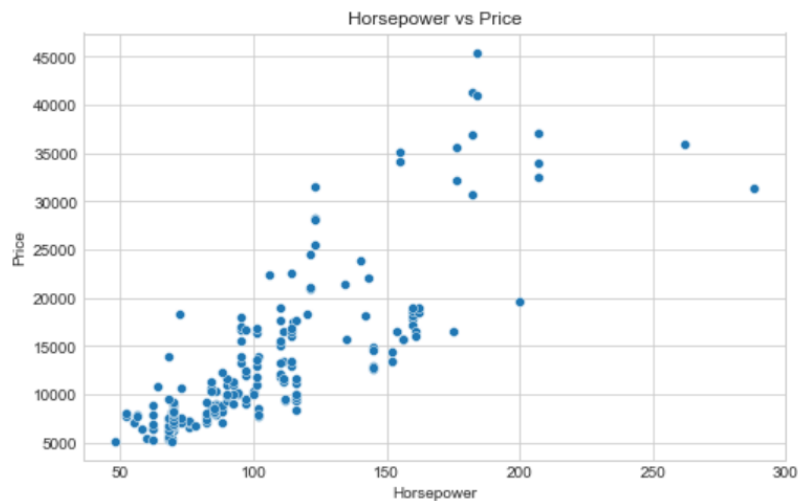
This chart compares prices of standard (std) vs. turbocharged (turbo) cars. Turbocharged cars are clearly more expensive. This is expected because turbo engines offer more power and performance.

7.2.4 Engine Size vs Price



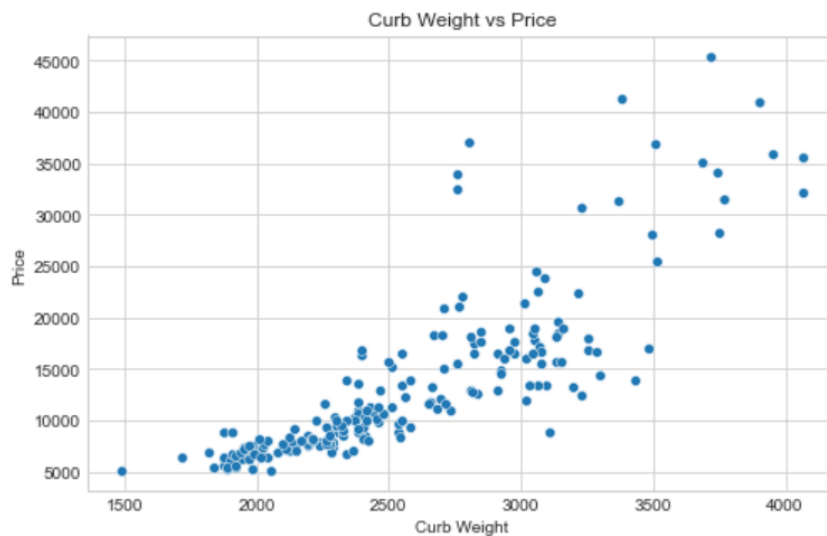
This scatter plot shows a positive relationship between engine size and price. As engine size increases, the price also increases. Cars with large engines (200–300) are usually premium or luxury vehicles.

7.2.5 Horsepower vs Price



This plot also shows a strong positive trend. Cars with higher horsepower tend to have higher prices. This makes sense because more horsepower usually means better performance.

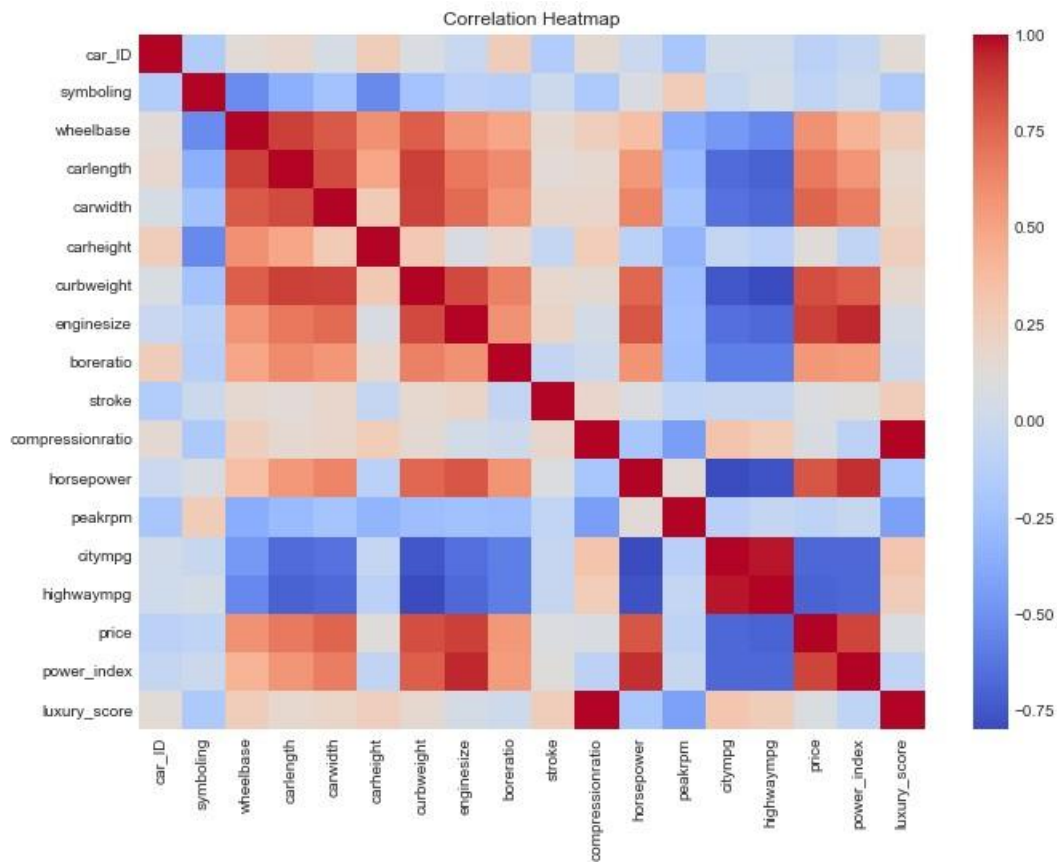
7.2.6 Curb Weight vs Price



This scatter plot shows that heavier cars generally cost more. Higher curb weight often indicates, better build quality, bigger engines and more safety features. All of these contribute to a higher price.

7.3 Multivariate Analysis

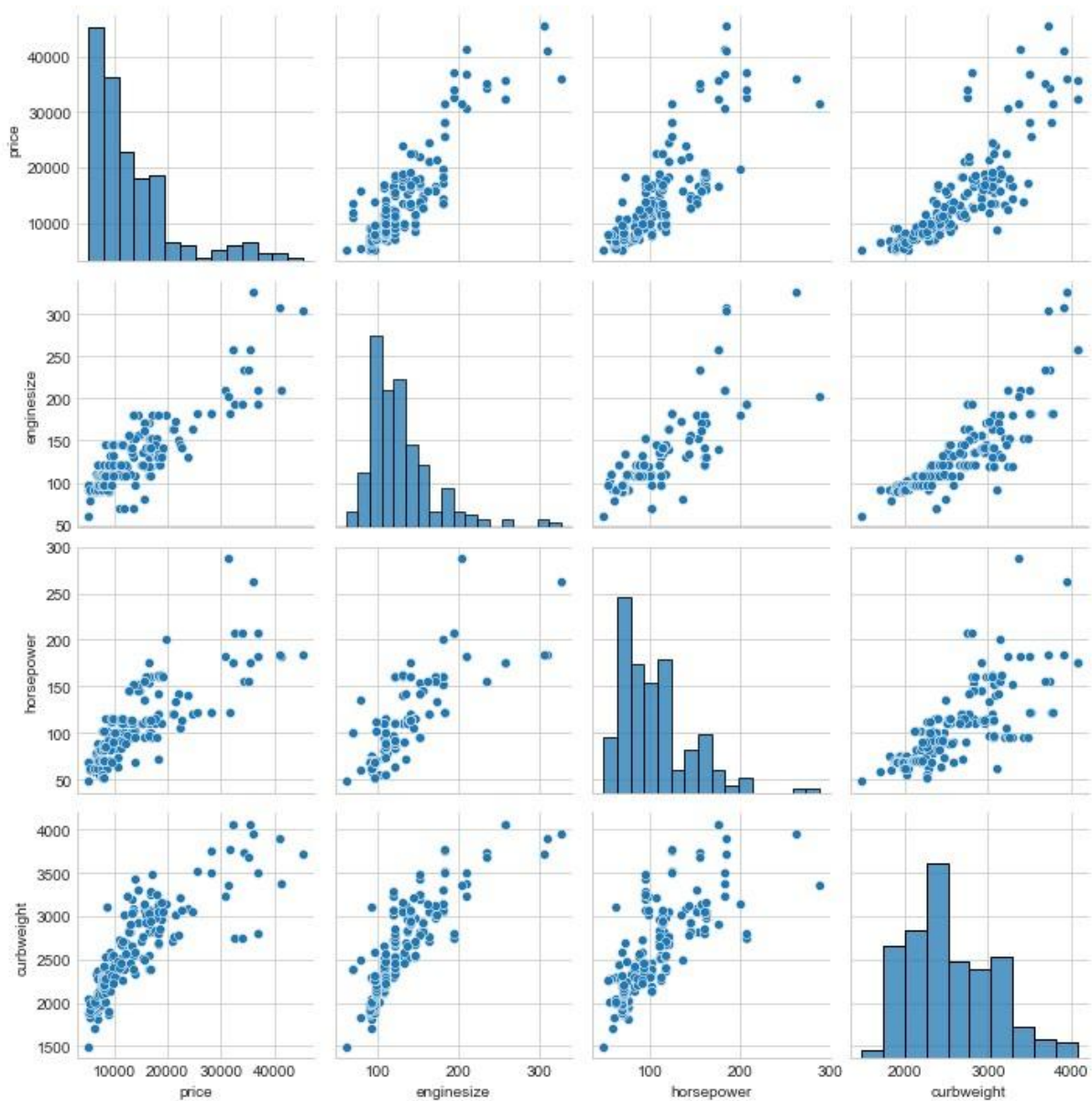
7.3.1 Correlation Heatmap



Features like engine size, horsepower, and curb weight show a strong positive correlation with price. When these values increase, the price usually increases too. City MPG and Highway MPG have a negative correlation with price. Expensive cars usually have lower mileage. Many engine-related features are related to each other because they describe similar parts of the car. The heatmap helps identify which features are most useful for prediction.

Therefore, Engine size, horsepower, curb weight, and power index are the strongest predictors of car price.

7.3.2 Pair Plot (Price, Engine Size, Horsepower, Curb Weight)



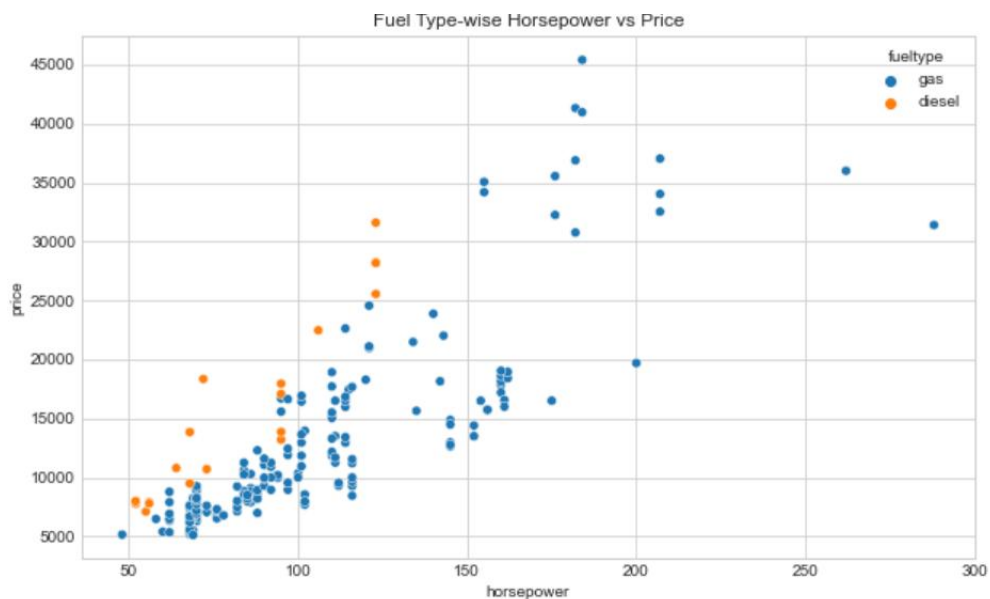
- ✓ As engine size increases, price also increases.
- ✓ Higher horsepower leads to higher price.
- ✓ Cars with higher curb weight (heavier cars) are also more expensive.
- ✓ The diagonal plots show distributions of each feature.

7.3.3 Brand-wise Engine Size vs Price



The plot shows that brands behave differently even for similar engine sizes. Luxury brands like BMW, Jaguar, Porsche have much higher prices even with similar engine sizes. Budget brands like Chevrolet, Dodge, Honda have lower prices. Therefore, brand modifies the relationship between engine size and price. Some brands charge more for the same engine size due to luxury value.

7.3.4 Fuel Type-wise Horsepower vs Price



Both diesel and gas cars show an upward trend → higher horsepower → higher price. Diesel cars (orange points) tend to be slightly more expensive than gas cars for the same horsepower level. High-horsepower gas cars still dominate the sporty/luxury segment. Therefore, fuel type changes the price pattern, even for the same horsepower.

Conclusion

The Exploratory Data Analysis (EDA) helped us clearly understand how different car features affect the price. From the analysis, we found that brand, engine size, horsepower, and curb weight are the strongest factors that increase the price. Luxury brands and cars with turbo engines or diesel fuel tend to be more expensive. We also saw that larger engines and higher horsepower cars usually cost more, while cars with high mileage (MPG) are generally cheaper. Overall, the EDA showed important patterns and relationships in the dataset, which helped us choose the right features and build better machine learning models.

8. Model Building

The main goal of this stage is to build machine learning models that can accurately predict the price of a car based on its features. After completing data cleaning, feature engineering, and exploratory data analysis, the prepared dataset was used to train several prediction models. To ensure reliable performance, the dataset was divided into training and testing sets, and numerical features were scaled to maintain uniformity across different variables. Multiple algorithms were applied to compare their accuracy and understand which model performs best for car price prediction. This model-building process helps identify patterns in the data and provides a strong foundation for selecting the most effective predictive approach.

After completing feature engineering, the dataset contained 205 rows and 69 features. The data was split into:

- 80% for training (164 rows)
- 20% for testing (41 rows)

8.1 Linear Regression

Only the numerical columns were standardized using StandardScaler so that all values fall within a similar range. This step is important for Linear Regression because it helps the model learn more effectively and prevents large-scale features from dominating the equations.

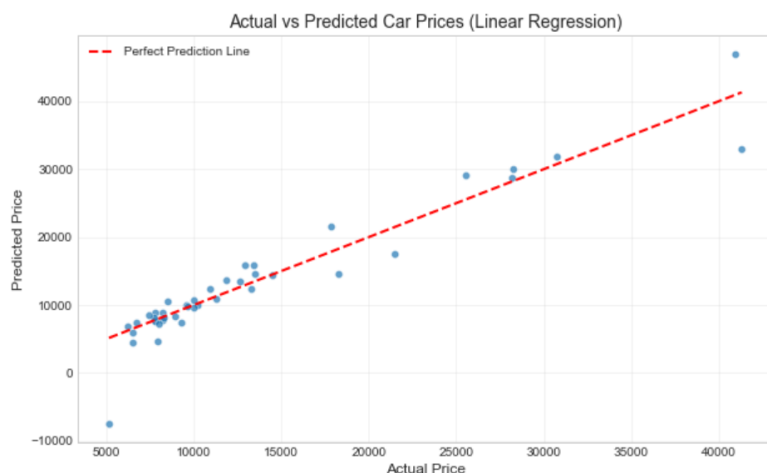
Model Performance

R² Score : 0.8832
MAE : 1837.84
RMSE : 3036.45

The Linear Regression model performs quite well, with an R² score of 0.88, meaning it can explain about 88% of the changes in car prices. The MAE and RMSE values show that, on average, the model's predictions are off by around \$1800 to \$3000. This means the model gives fairly accurate results for most cars, but it may not predict prices of very expensive luxury cars correctly because their price patterns are more complex and not purely linear.

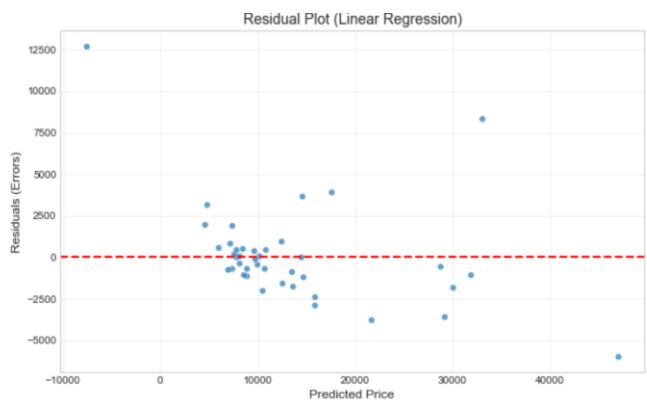
Actual vs Predicted Values and plot

	Actual Price	Predicted Price
0	30760.000	31834.000426
1	17859.167	21607.199712
2	9549.000	9964.585238
3	11850.000	13618.440840
4	28248.000	30031.992749
5	7799.000	7788.551660
6	7788.000	7580.577870
7	9258.000	7371.939350
8	10198.000	10087.471064
9	7775.000	8872.869578



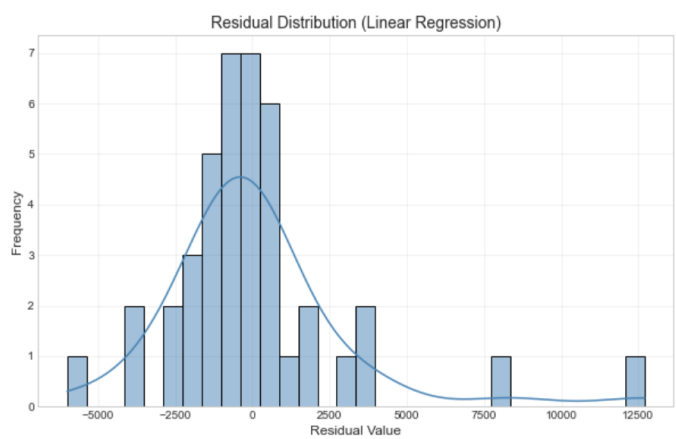
The values given in the table is for first 10 predictions and actual only. The plot shows how close the predicted car prices are to the actual prices. Most of the points lie near the diagonal line, indicating that the model's predictions are quite accurate. However, some points deviate from the line, especially for luxury car models, suggesting minor prediction errors in those cases.

Residual Plot



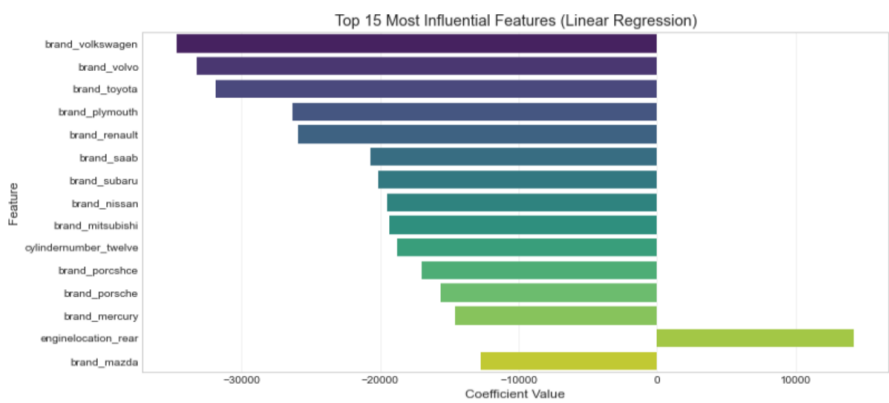
The plot displays the residuals, or errors, for each car price prediction. Most of the residuals are centered around zero, indicating that the model's predictions are generally accurate and consistent. However, a few large residuals reveal the instances where the model struggles to predict prices accurately.

Residual Distribution



The residuals roughly follow a normal distribution, suggesting that the model performs reasonably well overall. However, the presence of some outliers indicates that the model makes larger errors for certain observations.

Feature Importance



Some brands, such as Volkswagen, Volvo, and Toyota, have negative coefficients, indicating that they tend to lower the predicted car price. On the other hand, features like a rear-engine location and the Mazda brand contribute to higher predicted prices. Overall, the coefficients reveal which factors have the greatest impact on determining car prices.

8.2 Random Forest

The Random Forest model was trained using the same training dataset used for Linear Regression. Unlike Linear Regression, Random Forest does not require feature scaling, because decision-tree-based models work well with raw numerical values.

During training, the model built multiple decision trees and averaged their outputs to generate stable predictions. The default hyperparameters performed well, and no grid search tuning was applied.

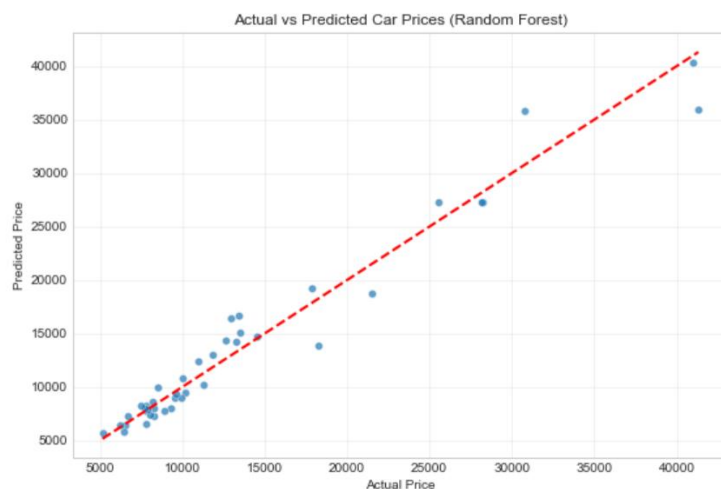
Model Performance

R^2 Score : 0.9579
MAE : 1287.29
RMSE : 1822.51

The R^2 score of 0.96 indicates that the model can explain 96% of the variation in car prices. The errors are much smaller compared to the Linear Regression model, showing improved accuracy. This suggests that the Random Forest model effectively captures the complex non-linear relationships present in the data.

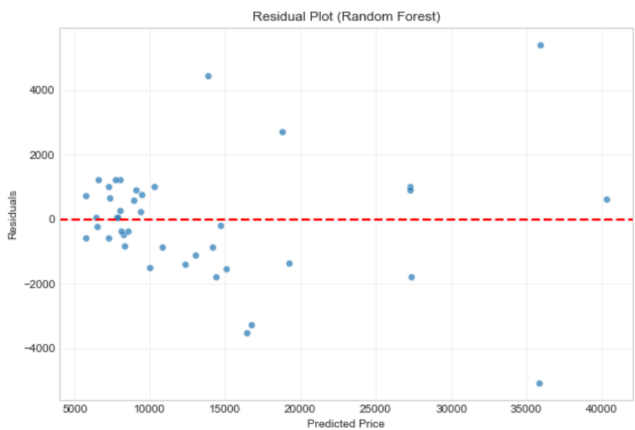
Actual vs Predicted Values and plot

	Actual	Predicted_RF
0	30760.000	35851.7275
1	17859.167	19205.5050
2	9549.000	8949.8600
3	11850.000	12979.4500
4	28248.000	27245.5975
5	7799.000	6585.3200
6	7788.000	7744.0550
7	9258.000	8038.4200
8	10198.000	9445.2400
9	7775.000	8266.7850



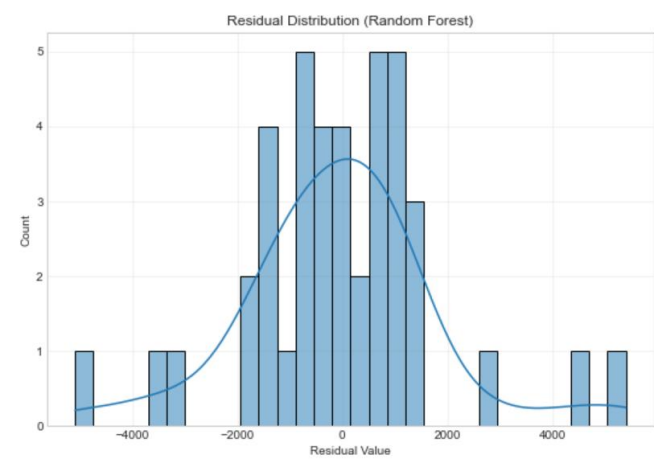
The points are closely aligned with the red diagonal line, indicating excellent prediction accuracy. The clustering is much tighter compared to the Linear Regression model, showing more consistent performance. Only a few deviations are observed for high-priced cars, which are generally more difficult to predict accurately.

Residual Plot



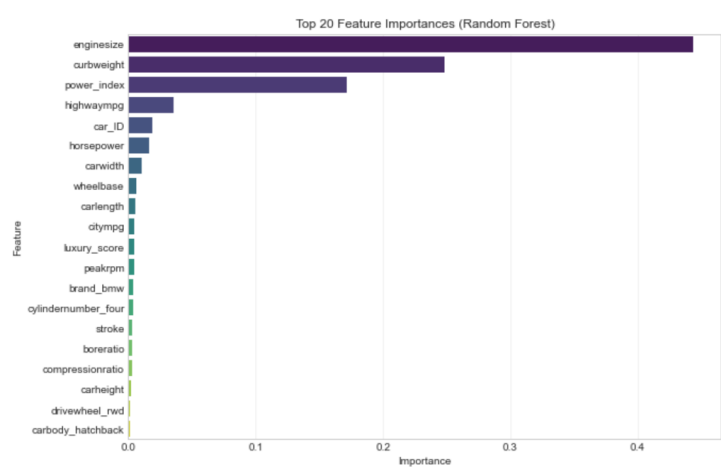
The residuals are mostly centered around zero, showing that the model’s predictions are generally accurate. Compared to Linear Regression, there are fewer large errors, indicating improved performance. The random distribution of residuals suggests that the model captures patterns well and is not affected by major bias.

Residual Distribution



Most residuals are concentrated around zero, indicating accurate predictions. The distribution is smoother and narrower than that of the Linear Regression model, showing greater consistency. This suggests that the Random Forest model makes smaller average errors and provides more reliable results.

Feature Importance



Engine-related features such as engine size, horsepower, curb weight, and power index have the greatest impact on car prices. While mileage also influences price, its effect is smaller compared to these performance factors. Brand features contribute less, as the Random Forest model emphasizes numerical variables. These results are consistent with the EDA findings, confirming that performance-related features are the key drivers of car prices.

8.3 XGBoost

The XGBoost Regression model was applied to further improve the prediction accuracy by capturing complex, non-linear relationships within the data. XGBoost is a boosting-based algorithm that builds trees sequentially, where each new tree attempts to correct the errors made by previous ones. This makes it highly effective for structured/tabular data like car specifications.

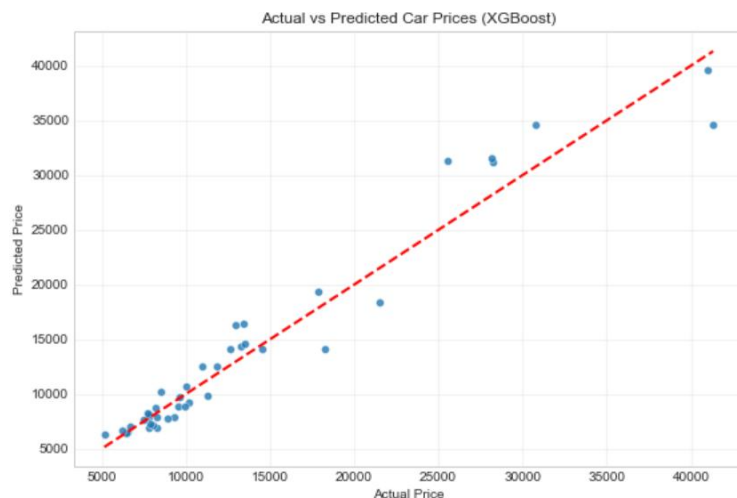
Model Performance

R² Score : 0.9416
MAE : 1524.05
RMSE : 2147.37
Best NTrees (best_iteration) : 81

The XGBoost model achieved an R² score of 0.94, meaning it explains about 94% of the variation in car prices. This reflects strong performance, placing it among the best models in the analysis, though slightly behind the Random Forest model. With an MAE of around 1524 and an RMSE of about 2147, the model demonstrates solid predictive accuracy and maintains relatively low error levels.

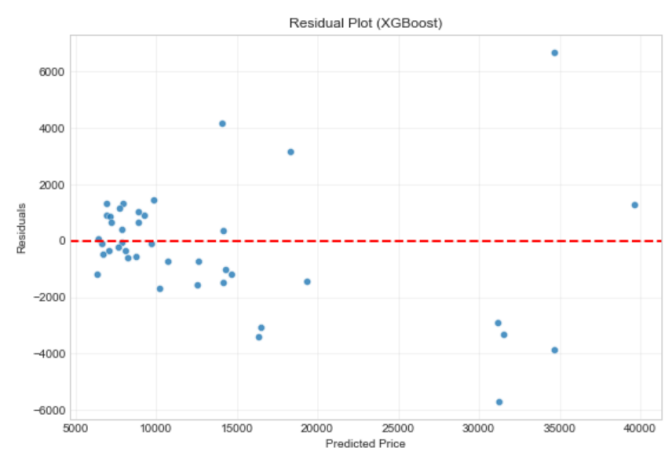
Actual vs Predicted Values and plot

	Actual	Predicted_XGB
0	30760.000	34632.261719
1	17859.167	19308.884766
2	9549.000	8903.475586
3	11850.000	12578.370117
4	28248.000	31132.111328
5	7799.000	6886.820312
6	7788.000	7855.729492
7	9258.000	7948.536621
8	10198.000	9282.956055
9	7775.000	8111.226074



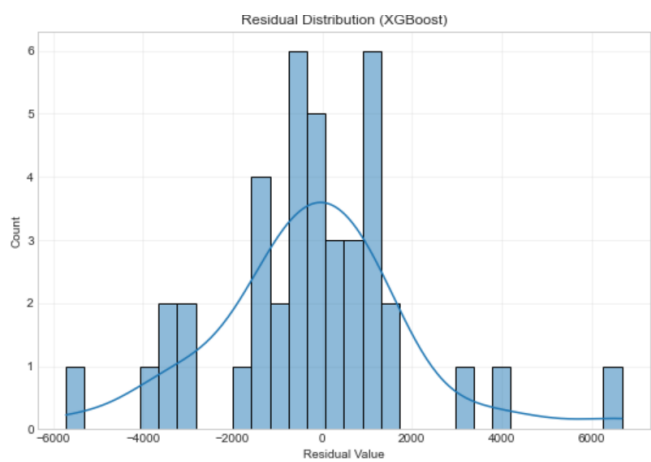
The Actual vs Predicted plot shows that most predicted values lie close to the ideal diagonal line, indicating that the model performs well for a majority of the cars. A few points deviate more noticeably, especially in the higher price range, suggesting that extremely expensive cars are slightly harder for the model to predict perfectly.

Residual Plot



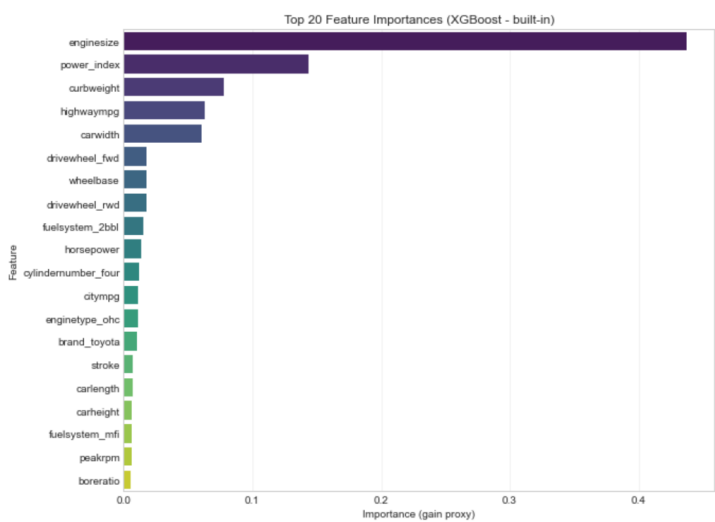
The residual plot also reflects this, with most residuals cluster The residuals are mostly centered around zero, showing that the model’s predictions are generally accurate. Compared to Linear Regression, there are fewer large errors, indicating improved performance. The random distribution of residuals suggests that the model captures patterns well and is not affected by major bias.

Residual Distribution



Most residuals are concentrated around zero, indicating accurate predictions. The distribution is smoother and narrower than that of the Linear Regression model, showing greater consistency. This suggests that the Random Forest model makes smaller average errors and provides more reliable results.

Feature Importance



The feature importance analysis reveals that engine size, power index, and curb weight are the top contributors to predicting price, confirming that performance-related features are the strongest price indicators.

9. Model Comparison

After building three different machine learning models Linear Regression, Random Forest Regressor and XGBoost Regressor their performances were evaluated and compared using the testing dataset. The goal of this comparison is to identify which model provides the most accurate and reliable car price predictions.

To ensure a fair comparison, the same train–test split, feature set, and evaluation metrics were used for all models.

- **R² Score:** shows how well the model explains the price variation
- **MAE (Mean Absolute Error):** average size of prediction mistakes
- **RMSE (Root Mean Squared Error):** gives more weight to large errors

	Model	R2 Score	MAE	RMSE
0	Linear Regression	0.883208	1837.836200	3036.449651
1	Random Forest	0.957925	1287.288366	1822.509713
2	XGBoost	0.941589	1524.052460	2147.367606

1. Linear Regression

Linear Regression did okay, but not great.

It got an R² of 0.883, which means it explains most of the variation, but:

- Its errors (MAE and RMSE) were higher than the other models.
- This shows that car prices don't follow a simple straight-line pattern.
- So, a basic linear model cannot fully understand the complex relationships in the data.

2. Random Forest Regressor

Random Forest performed the best among all models.

- **Highest R²:** 0.9579
- **Lowest MAE:** 1287
- **Lowest RMSE:** 1822

This means:

- It understands complex, non-linear patterns very well.
- It handles interactions between features better.
- It is strong even when the data has noise or irregularities.

Random Forest is the **most accurate and reliable model for predicting car prices** in this study.

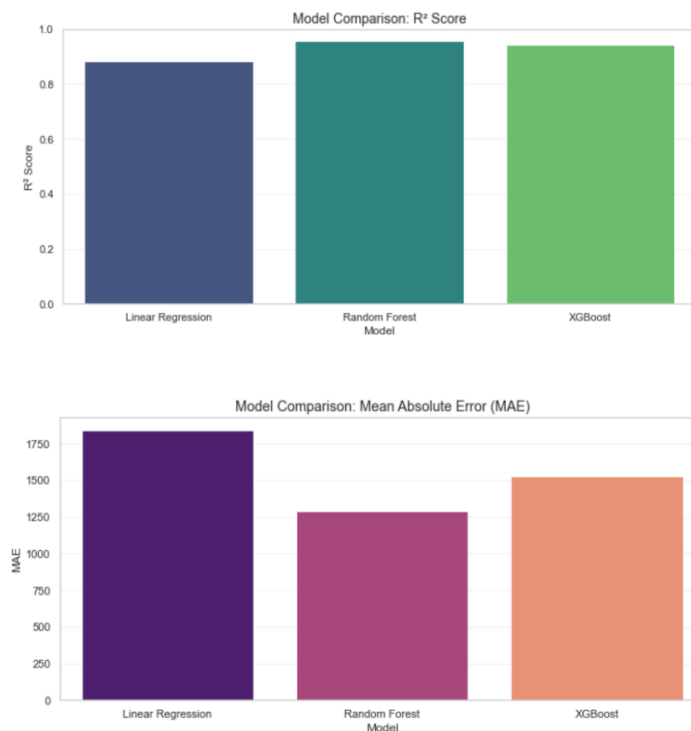
3. XGBoost Regressor

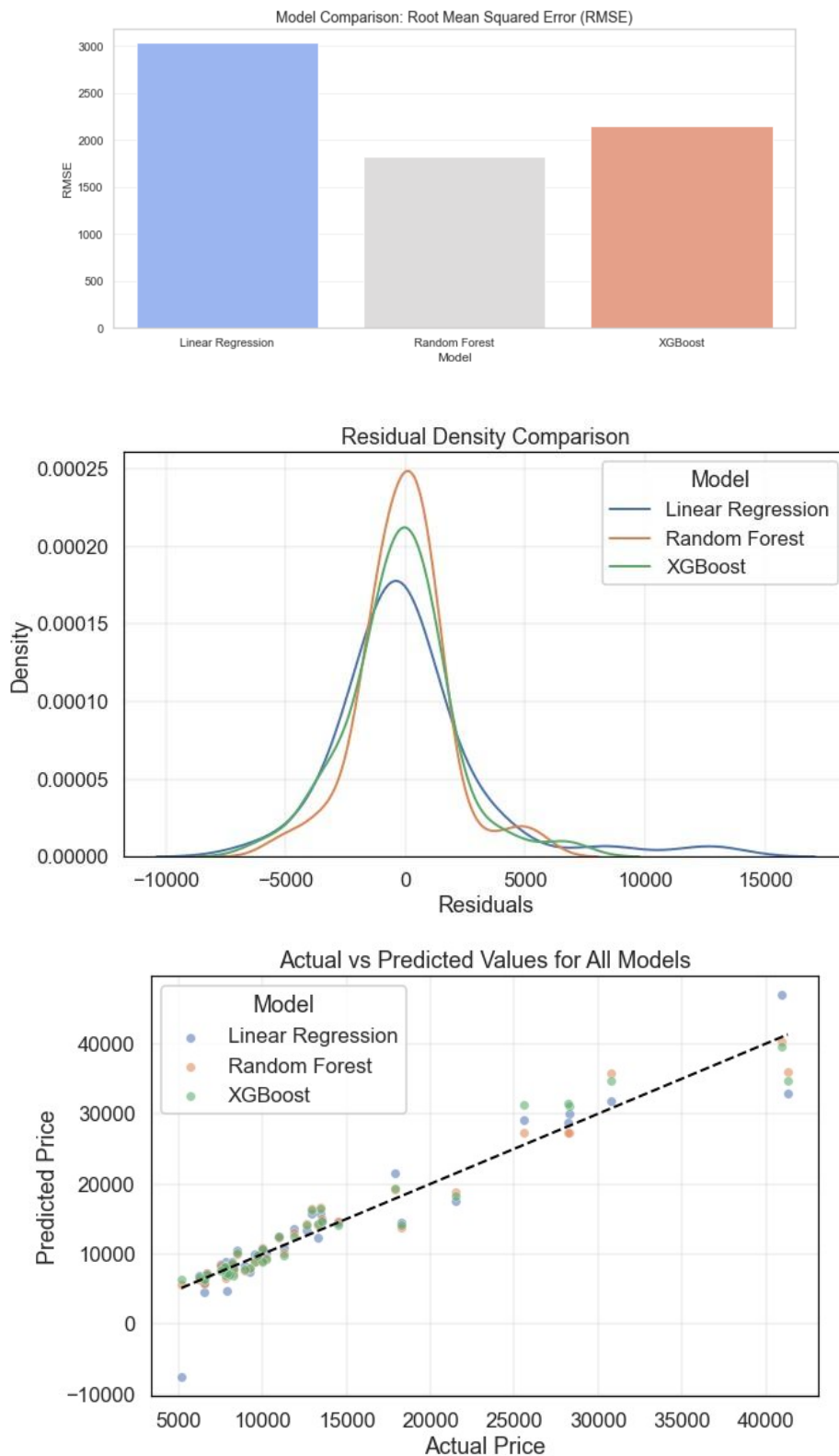
XGBoost also performed very well.

- R^2 (0.941) is better than Linear Regression.
- Errors are lower than Linear Regression, but slightly higher than Random Forest.

This shows the XGBoost is powerful and usually generalizes well. But in this dataset, Random Forest performed a little better.

Based on all evaluation metrics, **Random Forest is the best-performing model** for predicting car prices in this study. It provides the most accurate, consistent, and reliable results among the three models tested.





The **Residual Density Plot** shows that Random Forest has errors that are more tightly grouped around zero, meaning it makes fewer large mistakes. The **Actual vs Predicted Combined Plot** shows that points from all models follow the diagonal line, but Random Forest predictions lie closest to the perfect line, confirming it is the most accurate model.

10. Key Insights from the Best Model

The Random Forest model, which showed the strongest performance among the tested algorithms, provides several important insights into the factors that influence car pricing. The model's feature importance analysis highlights that **engine size**, **curb weight**, and **power index** are the most influential predictors of price. This means that cars with larger engines, heavier bodies, and higher power output tend to be priced significantly higher.

Horsepower also plays a major role, confirming that performance-oriented cars generally fall into the higher price range. Fuel efficiency (city and highway MPG) shows a moderate influence, suggesting that mileage affects price but is less impactful than engine-related specifications. Brand-related dummy variables appeared in the model but had lower importance, showing that while branding does influence price, the Random Forest model relied more heavily on **technical and performance-based characteristics** when making predictions.

Overall, the model reveals that **performance, size, and engineering specifications** are the key drivers of car pricing in this dataset.

11. Discussion / Key Findings

This study compared three machine learning models Linear Regression, Random Forest, and XGBoost to predict car prices, and several important findings emerged from the analysis:

- **Tree-based models significantly outperform Linear Regression:** Linear Regression achieved respectable results but struggled to capture the complex and non-linear relationships in the data. Random Forest and XGBoost delivered much better accuracy, showing that car price prediction requires models capable of handling non-linearity.
- **Random Forest produced the most accurate and stable predictions:** With the highest R^2 score and the lowest MAE and RMSE, Random Forest demonstrated strong generalization and handled both common and rare price ranges effectively.
- **XGBoost performed well but slightly below Random Forest:** Although XGBoost captured non-linear patterns effectively, its errors were slightly higher than Random Forest. This may be due to the relatively small dataset, where Random Forest tends to perform more consistently.

- **Key features influencing price were consistent across models:** Engine size, curb weight, horsepower, and power index repeatedly appeared as top predictors in all models. These features strongly correlate with a vehicle's performance and build quality, making them reliable indicators of price.
- **Residual and error plots confirm better stability in Random Forest:** The error distributions for Random Forest were tighter and more centered around zero, showing fewer large mistakes compared to the other models.

Overall, the analysis shows that **cars with larger, more powerful, and heavier engines are priced higher**, and **Random Forest provides the most accurate representation of these patterns**. The findings align with real-world pricing behaviour in the automotive market.

12. Conclusion

In this project, the goal was to predict car prices and find out which features affect the price the most. After cleaning the data and creating useful new features, we built and tested three models: Linear Regression, Random Forest, and XGBoost.

The results showed that car prices are mainly influenced by performance-related features like **engine size, horsepower, curb weight, and combined power index**.

Among all the models, Random Forest performed the best. It had the highest accuracy and the lowest error values. This means it was able to understand the complex, non-linear patterns in the data better than the other models.

Overall, the project successfully created a reliable model for predicting car prices and gave clear insights into what factors affect car pricing the most.